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Lesson 5: Operations with Fractions

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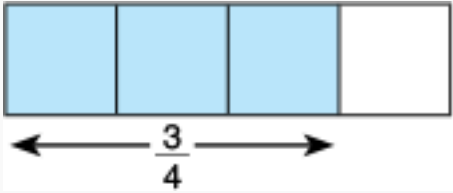
# Multiplying Fractions

When you multiply a fraction by a fraction, you are taking something small and breaking it into even smaller parts. You can expect the answer to be smaller than either of the fractions you started with.

Example 1

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Sylvia had  $\frac{3}{4}$  pound of candy. She wanted to give each of her 3 grandchildren  $\frac{1}{3}$  of the candy. How much candy should Sylvia give each grandchild?



Step 1

Multiply  $\frac{3}{4}$  by  $\frac{1}{3}$ . Look for ways to cancel (or factor out) a multiple common to the numerator and the denominator. This makes the problem simpler.

$\frac{3}{4} \times \frac{1}{3}$

Step 2

Cancel the numerator of the first fraction and the denominator of the second fraction.

$\frac{\cancel{3}^1}{4} \times \frac{1}{\cancel{3}_1}$

Step 3

Multiply across using the new numerators and denominators.

$\frac{1}{4} \times \frac{1}{1} = \frac{1}{4}$  pound

Example 2

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Multiply  $\frac{6}{7}$  by  $\frac{7}{9}$ .

Step 1

Look for ways to cancel. Notice that canceling may include both numerators and denominators.

$\frac{6}{7} \times \frac{7}{9}$

Step 2

Cancel the numerator of the first fraction and the denominator of the second fraction.

$\frac{\cancel{6}^2}{7} \times \frac{\cancel{7}_3}{9}$

Step 3

Cancel the numerator of the second fraction and the denominator of the first fraction.

$\frac{\cancel{2}^1}{\cancel{7}_3} \times \frac{\cancel{7}^1}{9}$

Step 4

Multiply across using the new numerators and denominators.

$\frac{2}{1} \times \frac{1}{3} = \frac{2}{3}$

If you did not cancel the fractions before multiplying, you would still get the same answer. However, the answer would have a larger numerator and denominator, and you would have to simplify your answer after multiplying. Therefore, it is much easier to cancel first.

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